Rayan Alam

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EDUCATION

University of California, Los Angeles **Expected June 2027**
M.S in Mechanical Engineering
Arizona State University
B.S. in Biomedical Engineering – *Cum Laude* **May 2025**
Dean's List: Fall 2021- Fall 2022, Spring 2024-Spring 2025

SKILLS

- **CAD & Design: 3D CAD** design (**SolidWorks**, Fusion 360, AutoCAD), parametric assembly design, **DFM**
- **Prototyping & Testing: prototyping**, design **iteration**, 3D printing, silicone molding, bench testing
- **Test & Validation: test execution**, **test data analysis**, Instron compression testing, validation against acceptance criteria, results documentation
- **Analysis & Documentation:** SolidWorks flow simulation, MATLAB, **technical reports**, **project documentation**, **Microsoft Office** (Word, Excel, PowerPoint)
- **Systems Integration: control system integration**, custom **PCB** for a safety cutoff switch, cross-functional research coordination

EXPERIENCE

Bionics Lab, UCLA- Research Assistant **September 2025 - June 2026**

- Designed a **custom PCB** to regulate a power/safety cutoff switch for the control system, enabling operators to immediately de-energize the teleoperation rig during testing and improving overall system safety.
- Assist in experimental setup, data collection, and **system integration** for human-subject studies.
- Supported **3** human-subject studies on VR-based rehabilitation by managing experimental setup and data collection, maintaining **95%** protocol compliance across sessions.
- Participate in research on medical and wearable robotics, with emphasis on biomechanics, control systems, and human-machine interaction.

Weaver Lab: Biomaterials and Immune Engineering-Research Assistant **August 2023 - May 2025**

- Designed and **iterated** device molds across **15** 3D-printed **prototypes** using **4** different print materials, identifying the optimal combination for dimensional accuracy through comparative testing.
- Performed **flow simulations** in SolidWorks to evaluate mass transport, informing geometry optimization that improved oxygen diffusion uniformity by **18%**.
- Achieved a consistent **50/50** material split for a Type 1 diabetes islet-encapsulation device by identifying **0.5 mL/min** as the optimal pump speed across dual-syringe co-extrusion of alginate and gluconic-delta-lactone (GDL) hydrogel trials (tested at **0.5, 1, and 3 mL/min**).

Massachusetts General Hospital (MGH) - Harvard Medical School-Research Assistant **June 2024 - August 2024**

- Restored colonic motor complex frequency and extended median survival from **10 to 26 days** by supporting translational research on enteric neural stem cell (ENSC) transplantation in a Hirschsprung disease mouse model.
- Achieved cell coverage of up to **16.8 mm²** via multi-injection laparotomy delivery, assisting in isolating, expanding, and microinjecting ENSCs as neurospheres into the aganglionic colon of Ednrb knockout mice.
- Conducted literature reviews and contributed to histological and motility analysis supporting the translational research program.

PROJECTS & RESEARCH

Implantable Leptin Delivery Pump-Prototype Lead **August 2024 - May 2025**

- Designed a full parametric **CAD assembly** in **SolidWorks** and selected device materials (Platinum Cure Silicone, ABS, SIL-Poxy), including a custom silicone bellow for controlled, low-flow drug delivery; applied **design-for-manufacturability (DFM)** practices to address low-volume silicone molding and tooling constraints.
- Validated durability via **Instron compression testing** across **10-500 kPa** (**15 total pressure cycles**), achieving structural integrity scores of **9-10 out of 10** with zero leaks or failures — exceeding the project's **≥8/10** reusability threshold.
- Held a **0.3 mL/day** leptin dosing profile within a **0.2-0.4 mL/day** acceptance range across three physiological temperatures (**97-102°F**) over **4-day** trials, repeated across **3 independent runs**, engineering the pump for cerebrospinal fluid delivery to prevent Type II diabetes.

CleanWater Toilets-Prototype Designer **August 2021 -December 2021**

- Cut assembly time **40%** for non-technical stakeholders by redesigning a bag-and-frame toilet system into a single-step snap-fit Fusion 360 CAD assembly.
- Designed a low-cost, bag-based dispensable toilet system addressing open defecation for an estimated **154 million** people in rural India, and pitched a full business plan sizing the addressable market at **34 million** underserved rural households with **5%** projected penetration over **3 years**.